Mini Project

Banking Operations Management System Using C# Console Application, ADO.NET and SQL Server

Develop a **Banking Operations Management System** using:

- C# Console Application
- ADO.NET
- SQL Server
- Stored Procedures

The application should simulate real-world banking operations with two modules:

1. Admin Module
2. Customer Module

Objective

The objective of this project is to understand and implement:

- Console-based application development
- ADO.NET connectivity
- SQL Server database operations
- Stored Procedures
- CRUD operations
- Authentication and Login
- Transaction handling
- Menu-driven applications
- Layered programming approach

Project Modules

Module 1: Admin

The Admin should be able to:

- Login securely
- Add new customers
- Update customer details
- Delete customers
- View all customers

- Search customer by account number
- View customers by account type
- View all transactions
- Freeze or activate customer account

Module 2: Customer

The Customer should be able to:

- Register new account
- Login securely
- Deposit money
- Withdraw money
- Check account balance
- View mini statement
- Update profile details
- Change password
- Transfer funds to another account
- Close account

Database Design

Create a database named:

BankDB

Required Tables

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1. Customers Table

Create a table named:

Customers

Suggested Columns

Column Name	Data Type
CustomerId	INT (Primary Key, Identity)
FullName	VARCHAR
Gender	VARCHAR
MobileNumber	VARCHAR
Email	VARCHAR
Address	VARCHAR
AadhaarNumber	VARCHAR
AccountType	VARCHAR
AccountNumber	BIGINT
Password	VARCHAR
Balance	DECIMAL
AccountStatus	VARCHAR
CreatedDate	DATETIME

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2. Transactions Table

Create a table named:

Transactions

Suggested Columns

Column Name	Data Type
TransactionId	INT (Primary Key, Identity)
AccountNumber	BIGINT
TransactionType	VARCHAR
Amount	DECIMAL
TransactionDate	DATETIME
BalanceAfterTransaction	DECIMAL

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3. Admin Table

Create a table named:

Admins

Suggested Columns

Column Name	Data Type
AdminId	INT
UserName	VARCHAR
Password	VARCHAR

Functional Requirements

Admin Functionalities

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1. Admin Login

Admin should login using:

- Username
- Password

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2. Add Customer

Admin can:

- Create customer account
- Assign account number
- Set initial balance
- Select account type:
    - Savings
    - Current
    - Salary

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3. Update Customer

Admin can modify:

- Mobile Number
- Email
- Address
- Account Type
- Account Status

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4. Delete Customer

Admin can delete customer records.

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5. View All Customers

Display:

- Customer Id
- Name
- Account Number
- Account Type
- Balance
- Status

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6. Search Customer

Search using:

- Account Number
- Mobile Number

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7. View Customers by Account Type

Example:

1. Savings
2. Current
3. Salary

Display all customers belonging to selected type.

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8. View Transaction History

Admin can see all banking transactions.

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9. Freeze/Activate Account

Admin can:

- Freeze suspicious account
- Reactivate account

Customer Functionalities

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1. Register

Customer should provide:

- Name
- Mobile Number
- Address
- Aadhaar Number
- Account Type
- Password

System should generate:

Account Number

automatically.

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2. Customer Login

Login using:

- Account Number
- Password

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3. Deposit Money

Customer can:

- Enter amount
- Update account balance
- Store transaction history

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4. Withdraw Money

Conditions:

- Withdrawal amount should not exceed balance
- Minimum balance rules can be applied

Store transaction history after withdrawal.

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5. Show Balance

Display current account balance.

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6. Mini Statement

Display last 5 or 10 transactions.

Example:

Date Type Amount Balance

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7. Fund Transfer

Transfer money between accounts.

Requirements:

- Validate receiver account
- Deduct sender balance
- Add receiver balance
- Store transaction history

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8. Change Password

Customer should update password securely.

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9. Update Profile

Customer can update:

- Mobile Number
- Email
- Address

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10. Close Account

Customer can request account closure.

Stored Procedure Requirements

Students must create stored procedures for:

Operation	Procedure Example
Customer Registration	sp_RegisterCustomer
Customer Login	sp_CustomerLogin
Admin Login	sp_AdminLogin
Deposit	sp_Deposit
Withdraw	sp_Withdraw
Balance Check	sp_GetBalance
Mini Statement	sp_GetMiniStatement
Fund Transfer	sp_TransferFunds
Add Customer	sp_AddCustomer
Update Customer	sp_UpdateCustomer
Delete Customer	sp_DeleteCustomer
Get Customers by Type	sp_GetCustomersByType

ADO.NET Requirements

Students must use:

- SqlConnection
- SqlCommand
- SqlDataReader
- SqlTransaction
- ExecuteReader()
- ExecuteScalar()
- ExecuteNonQuery()

Console Application Requirements

The application should be:

- Menu driven
- User friendly
- Modular
- Exception handled

Suggested Console Menus

Main Menu

1. Admin Login
2. Customer Login
3. Customer Registration
4. Exit

Admin Menu

1. Add Customer
2. Update Customer
3. Delete Customer
4. View All Customers
5. Search Customer
6. View Customers by Account Type
7. View Transactions
8. Freeze Account
9. Logout

Customer Menu

1. Deposit
2. Withdraw
3. Show Balance
4. Mini Statement
5. Fund Transfer
6. Update Profile
7. Change Password
8. Close Account
9. Logout

Guidelines

Database Guidelines

- Use SQL Server
- Use Stored Procedures only
- Normalize tables properly
- Use proper constraints and primary keys

Coding Guidelines

- Follow naming conventions
- Use layered programming approach
- Use methods for reusable operations
- Use exception handling
- Use transactions for fund transfer

Security Guidelines

- Validate login credentials
- Avoid SQL Injection using parameters
- Validate balance before withdrawal
- Use proper password validations

Transaction Guidelines

For:

- Fund Transfer
- Deposit
- Withdraw

Students should use:

SqlTransaction

to maintain data consistency.

Expected Learning Outcomes

After completing this project, students should be able to:

- Develop menu-driven console applications
- Work with SQL Server databases
- Perform CRUD operations using ADO.NET
- Create and use Stored Procedures
- Implement authentication systems
- Handle banking transactions
- Use transactions in SQL Server
- Design modular applications

Optional Enhancements

Students can additionally implement:

1. OTP Verification
2. Email Notifications
3. Interest Calculation
4. Loan Management
5. ATM Simulation
6. PIN Generation
7. PDF Statement Generation
8. Role-Based Authorization
9. Account Lock after Multiple Failed Attempts
10. Dashboard Reports

Deliverables

Students should submit:

1. SQL Scripts
2. Console Application Source Code
3. Stored Procedures
4. Output Screenshots
5. Project Documentation
6. ER Diagram
7. Test Cases